

No SAE J1939 Communication Remote Panel

Symptoms

- No SAE J1939 Communication with the remote panel instrument panel.
- Engine room panel has SAE J1939 communication.

How To Use This Tree

This symptom tree can be used to troubleshoot SAE J1939 communications symptoms.

Start by checking the terminating resistors. There are two terminating resistors. The terminating resistors are located at the following points:

One resistor is located on the engine wiring harness.

The other resistor is located at the last remote panel, at the X4 terminal strip between the SAE J1939 Supply and SAE J1939 Return terminals.

Step 1 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The SAE J1939 datalink provides information to the instrument panel in the remote panel.

The SAE J1939 datalink provides the following parameters:

- Engine fault codes
- Engine parameters monitored by the ECM.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check SAE J1939 Datalink Signal	
	STEP 1A. Check SAE J1939 Datalink Communication on Engine	Communication established?
	STEP 1B. Check SAE J1939 Datalink Communication at Remote Panel	Communication established?

STEPS		SPECIFICATIONS
STEP 2.	Check Remote Panel Wiring	
	STEP 2A. Check Remote Panel SAE J1939 Datalink Supply Wire (Instrument Panel Connector to Service Port Connector)	Less than 10 ohms resistance?
	STEP 2B. Check Remote Panel SAE J1939 Datalink Return Wire (Instrument Panel Connector to Service Port Connector)	Less than 10 ohms resistance?
	STEP 2C. Check Remote Panel SAE J1939 Datalink Shield Wire (Instrument Panel Connector to Service Port Connector)	Less than 10 ohms resistance?
STEP 3.	Check Panel System Cables	
	STEP 3A. Check Remote Panel Cable (SAE J1939 Supply and Return Wires)	Less than 10 ohms resistance?
	STEP 3B. Check Remote Panel Cable (SAE J1939 Return and Shield Wires)	Less than 10 ohms resistance?
	STEP 3C. Check Engine Room Panel Cable (SAE J1939 Supply and Return Wires)	Less than 10 ohms resistance?
	STEP 3D. Check Engine Room Panel Cable (SAE J1939 Return and Shield Wires)	Less than 10 ohms resistance?
STEP 4.	Check Customer Interface Box Wiring	
	STEP 4A. Check SAE J1939 Datalink Supply Wire	Less than 10 ohms resistance?
	STEP 4B. Check SAE J1939 Datalink Return Wire	Less than 10 ohms resistance?
	STEP 4C. Check SAE J1939 Datalink Shield Wire	Less than 10 ohms resistance?
STEP 5.	Check Engine Room Panel Wiring	

STEPS		SPECIFICATIONS
	STEP 5A. Check SAE J1939 Datalink Supply Wire (C14 Connector to Instrument Panel Connector)	Less than 10 ohms resistance?
	STEP 5B. Check SAE J1939 Datalink Return Wire (C14 Connector to Instrument Panel Connector)	Less than 10 ohms resistance?
	STEP 5C. Check SAE J1939 Datalink Shield Wire (C14 Connector to Instrument Panel Connector)	Less than 10 ohms resistance?
STEP 6.	Check Display Wiring	
	STEP 6A. Check Remote Panel SAE J1939 Datalink Supply Wire (Instrument Panel Connector to Display)	Less than 10 ohms resistance?
	STEP 6B. Check Remote Panel SAE J1939 Datalink Return Wire (Instrument Panel Connector to Display)	Less than 10 ohms resistance?

STEP 1. Check SAE J1939 Datalink Signal

STEP 1A. Check SAE J1939 Datalink Communication on Engine

Conditions :

- Locate engine wiring harness
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Check SAE J1939 datalink communication. Use INSITE™ electronic service tool to establish communication.	Communication established? YES	1B
	Communication established? NO Repair : Refer to the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html).	Repair complete.

STEP 1B. Check SAE J1939 Datalink Communication at Remote Panel

Conditions :

- Locate remote panel
- Connect electronic service tool.

Action	Specification/Repair	Next Step
Check SAE J1939 datalink communications. Use INSITE™ electronic service tool to establish communication.	Communication established? YES	6A
	Communication established? NO	2A

STEP 2. Check Remote Panel Wiring

STEP 2A. Check Remote Panel SAE J1939 Datalink Supply Wire (Instrument Panel Connector to Service Port Connector)

Conditions : Locate remote panel.		
Action	Specification/Repair	Next Step
Check remote panel wiring. - Disconnect remote panel cable from the instrument panel X4 connector. - Connect one test lead on the remote panel SAE J1939 datalink supply pin on the instrument panel X4 connector. - Place the other test lead on the remote panel SAE J1939 datalink supply pin on the service port connector.	Less than 10 ohms resistance? YES	2B
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	1B

STEP 2B. Check Remote Panel SAE J1939 Datalink Return Wire (Instrument Panel Connector to Service Port Connector)		
Conditions : Locate remote panel.		
Action	Specification/Repair	Next Step
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Check remote panel wiring. • Disconnect remote panel cable from the instrument panel connector. • Connect one test lead on the remote panel SAE J1939 datalink return pin on the instrument panel connector. • Place the other test lead on the remote panel SAE J1939 datalink return pin on the service port connector.	Less than 10 ohms resistance? YES	2C
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	1B

STEP 2C. Check Remote Panel SAE J1939 Datalink Shield Wire (Instrument Panel Connector to Service Port Connector)

Conditions : • Locate remote panel		
Action	Specification/Repair	Next Step
Check remote panel wiring. • Disconnect remote panel cable from the instrument panel connector. • Connect one test lead on the remote panel SAE J1939 datalink shield pin on the instrument panel connector. • Place the other test lead on the remote panel SAE J1939 datalink shield pin on the service port connector.	Less than 10 ohms resistance? YES	3A
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	1B

STEP 3. Check Panel System Cables

STEP 3A. Check Remote Panel Cable (SAE J1939 Supply and Return Wires)

Conditions :

- Locate and open customer interface box
- Locate remote panel.

Action	Specification/Repair	Next Step
Check the remote panel cable. • Install a jumper between the remote panel SAE J1939 datalink supply terminal and the remote panel SAE J1939 datalink return terminal on instrument panel X4 in the remote panel. • Place one test lead in the SAE J1939 datalink supply terminal in customer interface box X4 connector. • Place the other test lead in the SAE J1939 datalink return terminal in the customer interface box X4 connector.	Less than 10 ohms resistance? YES	3B
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 3B. Check Remote Panel Cable (SAE J1939 Return and Shield Wires)

Conditions :

- Locate and open customer interface box
- Locate remote panel.

Action	Specification/Repair	Next Step
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Check the remote panel cable. • Install a jumper between remote panel SAE J1939 datalink return terminal and the remote panel SAE J1939 datalink shield terminal on instrument panel X4 in the remote panel. • Place one test lead on the SAE J1939 datalink return terminal in customer interface box X4 connector. • Place the other test lead on the SAE J1939 datalink shield terminal in the customer interface box X4 connector.	Less than 10 ohms resistance? YES	3C
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 3C. Check Engine Room Panel Cable (SAE J1939 Supply and Return Wires)

Conditions :

- Disconnect cable connector C14 from the engine room panel.
- Disconnect cable connector C7 from the customer interface box.

Action	Specification/Repair	Next Step
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Check the engine room panel cable. • Install a jumper between engine room panel SAE J1939 datalink supply pin and the engine room panel SAE J1939 datalink return pin in connector C14. • Place one test lead in the engine room panel SAE J1939 datalink supply pin in connector C7. • Place the other test lead in the engine room panel SAE J1939 datalink return pin in connector C7.	Less than 10 ohms resistance? YES	3D
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 3D. Check Engine Room Panel Cable (SAE J1939 Return and Shield Wires)

Conditions :

- Disconnect cable connector C14 from the engine room panel.
- Disconnect cable connector C7 from the customer interface box.

Action	Specification/Repair	Next Step
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Check the engine room panel cable. • Install a jumper between engine room panel SAE J1939 datalink return pin and the engine room panel SAE J1939 datalink shield pin in connector C14. • Place one test lead in the engine room panel SAE J1939 datalink return pin in connector C7. • Place the other test lead in the engine room panel SAE J1939 datalink shield pin in connector C7.	Less than 10 ohms resistance? YES	4A
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 4. Check Customer Interface Box Wiring

STEP 4A. Check SAE J1939 Datalink Supply Wire

Conditions :

- Open the customer interface box
- Disconnect engine room cable at connector C7 from the customer interface box.

Action	Specification/Repair	Next Step

Check the SAE J1939 datalink supply wire. - Place one test lead on the SAE J1939 datalink supply (C3) pin in connector C7. - Place the other test lead on the engine SAE J1939 datalink supply terminal in the customer interface box X4 connector.	Less than 10 ohms resistance? YES	4B
	Less than 10 ohms resistance? NO Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 4B. Check SAE J1939 Datalink Return Wire

Conditions : - Open the customer interface box - Disconnect engine room cable at connector C7 from the customer interface box.		
Action	Specification/Repair	Next Step
Check the SAE J1939 datalink return wire. - Place one test lead on the SAE J1939 datalink return (C3) pin in connector C7. - Place the other test lead on the SAE J1939 datalink return terminal in the customer interface box X4 connector.	Less than 10 ohms resistance? YES	4C
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 4C. Check SAE J1939 Datalink Shield Wire

Conditions :

- Open the customer interface box
- Disconnect engine room cable at connector C7 from the customer interface box.

Action	Specification/Repair	Next Step
Check the SAE J1939 datalink shield wire. • Place one test lead on the SAE J1939 datalink shield (C3) pin in connector C7. • Place the other test lead on the SAE J1939 datalink shield terminal in the customer interface box X4 connector.	Less than 10 ohms resistance? YES	5A
	Less than 10 ohms resistance? NO Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 5. Check Engine Room Panel Wiring

STEP 5A. Check SAE J1939 Datalink Supply Wire (C14 Connector to Instrument Panel Connector)

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
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Check wires between harness connector and instrument panel connector. • Disconnect cable C14 from the engine room panel. • Place one test lead on the engine room panel SAE J1939 datalink supply pin on connector C14. • Place the other test lead on the engine room panel SAE J1939 datalink supply pin on the instrument panel connector.	Less than 10 ohms resistance? YES	5B
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 5B. Check SAE J1939 Datalink Return Wire (C14 Connector to Instrument Panel Connector)

Conditions : • Locate engine room panel • Open engine room panel door.		
Action	Specification/Repair	Next Step

Check wires between harness connector and instrument panel connector. • Disconnect cable C14 from the engine room panel. • Place one test lead on the engine room panel SAE J1939 datalink return pin on connector C14. • Place the other test lead on the engine room panel SAE J1939 datalink return pin on the instrument panel connector.	Less than 10 ohms resistance? YES	5C
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 5C. Check SAE J1939 Datalink Shield Wire (C14 Connector to Instrument Panel Connector)

Conditions : • Locate engine room panel • Open engine room panel door.		
Action	Specification/Repair	Next Step

Check wires between harness connector and instrument panel connector. • Disconnect cable C14 from the engine room panel. • Place one test lead on the engine room panel SAE J1939 datalink shield pin on connector C14. • Place the other test lead on the engine room panel SAE J1939 datalink shield pin on the instrument panel connector.	Less than 10 ohms resistance? YES	6A
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 6. Check Display Wiring

STEP 6A. Check Remote Panel SAE J1939 Datalink Supply Wire (Instrument Panel Connector to Display)

Conditions :

- Locate remote panel.

Action	Specification/Repair	Next Step

Check remote panel wiring. • Disconnect remote panel cable from the instrument panel connector. • Place one test lead on the remote panel SAE J1939 datalink supply pin on the instrument panel connector. • Place the other test lead on the remote panel SAE J1939 datalink supply wire at the display.	Less than 10 ohms resistance? YES	6B
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.

STEP 6B. Check Remote Panel SAE J1939 Datalink Return Wire (Instrument Panel Connector to Display)

Conditions :

- Locate engine room panel.

Action	Specification/Repair	Next Step
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Check remote panel wiring. • Disconnect remote panel cable from the instrument panel connector. • Place one test lead on the remote panel SAE J1939 datalink return pin on the instrument panel connector. • Place the other test lead on the remote panel SAE J1939 datalink return wire at the display.	Less than 10 ohms resistance? YES Repair : Replace the display. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.